Cart lab: change mass

antocciphysics | TEACHER KEY | CPA

Learning target: Use a controlled comparison to describe the effect of system mass on acceleration.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 8; 3 course days later.*
a=3 m/s² east; average velocity 6 m/s east; displacement 24 m east.

02. Explain a worked example (worked)
Points (1, 4),(2, 2); constant net force 4 N. Two points alone do not establish every detail of a curve.

03. Read the sample pattern (model)
Acceleration halves from 4 to 2 m/s² for the same 4 N net force.

04. Plan a fair test (practice)
Change total mass by adding to cart; measure acceleration; keep hanger mass/track/release same. More hanger changes driving force too.

05. Record evidence honestly (reason)
Teacher checks measured values and repeatability if collected. Supplied comparison 1 kg→2 kg: 4→2 m/s². Never claim supplied numbers are measured.

06. Choose a method (mixed) *Review: Unit 1, Day 10; 1 course days later.*
Earth pulls the sign down with 40 N; the cord pulls it up with 40 N.

07. Exit check (exit)
3 m/s²; same net force, comparable resistance. In real data deviations require checking controls.

08. STRETCH: Apply it another way (stretch)
1 m/s²; keep driving/friction conditions controlled, measure actual acceleration and compare including uncertainty.

Teaching note: Lab companion. Use the teacher-approved apparatus and release procedure. Supplied tables are ideal model data; record actual observations separately.

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Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 8; 3 course days later.*
Stops after 6 s at distance 72 m. Extending to 8 s predicts -8 m/s, which assumes continued acceleration after stopping and reversal.

02. Explain a worked example (worked)
Points (1, 4),(2, 2); constant net force 4 N. Two points alone do not establish every detail of a curve.

03. Test an inverse relationship (model)
1/M=2, 1.333, 1, 0.667 kg^{-1}; straight through origin with slope≈0.49 N, the net force.

04. System boundary (practice)
Hanger accelerates with cart; tension forces between components cancel in combined system. Idealizations massless string/pulley, negligible friction, inextensible string.

05. Measured slope and repetitions (reason)
a=.30/.60=.50 m/s^2. Table includes M, 3 trials, mean a, 1/M with units. No invented measurement accepted.

06. Choose a method (mixed) *Review: Unit 1, Day 10; 1 course days later.*
Weight 29.4 N down from Earth and tension 29.4 N up from string. Object-on-string acts on the string, not the object.

07. Exit check (exit)
F=.48 N; a=.40 m/s^2. Greater resistance lowers net force and observed acceleration.

08. STRETCH: Apply it another way (stretch)
Accelerations 6, 3, 1.5 m/s^2. Plot a versus 1/m; slope 3 N. Net force must be controlled, not assumed from a drive setting alone.

Teaching note: Lab companion. Use the teacher-approved apparatus and release procedure. Supplied tables are ideal model data; record actual observations separately.